

Referee Report, May 2012

This paper uses an impressive administrative data set from Finland to estimate the effect of job displacements on fertility. The sample includes the universe of residents (of which they use a 33% random sample) and allows the authors to identify mass layoffs and plant closures. Unlike earlier studies, this paper estimates both the effect of male and female job losses in the same context, which is a nice contribution. Estimated effects are based on a regression model that includes a series of indicator variables for leads and lags of job displacement in addition to a large set of control variables.¹ The authors focus on displacements taking place between 1991 and 1993, which allows them to observe outcomes 4 years before and 11-13 years following displacement. After documenting the effects on employment/earnings/marriage/etc., the authors estimate the effects on the probability of having a child in a given year and the cumulative number of children in a given year. These estimates provide suggestive evidence that a woman's displacement decreases fertility while male displacement has no impact on fertility (tables 3 and 4) though this is not apparent in the graphical analysis (figures 5 and 6). My major comments and concerns are as follows:

Are Displacements Exogenous?

Though the authors show that displaced and non-displaced workers are similar along most characteristics in tables 1 and 2, the fact that the authors often find significant “effects” of displacement prior to the displacement event is very concerning. Of particular concern is the fact that the authors almost always find a significant “effect” on the probability of being employed three years prior to displacement. This is pretty clear evidence of selection. In particular, this is evidence that individuals who tend to be more likely to be employed are more likely to be displaced. The authors might be able to address this problem by focusing on high-tenure workers.

In addition, when focusing on plant closures (which are more plausibly exogenous than mass layoffs) the authors find a significant negative “effect” on the probability of having a child three years before the displacement. Though it would not be inconceivable for this to be the case *ex ante*, since the effects of working in a distressed firm are often manifested in this way in other papers, in this case it is not plausible. The authors do not find that economic outcomes depart from their expected levels prior to displacements. Moreover, the estimated impact on the probability of having a birth 3 years prior to displacement is as big as any other estimate.

¹ Controls include age at displacement, age squared, a dummy for education level (6 categories), a dummy for education field (10 categories), pre-displacement years of tenure, tenure squared, pre-displacement marital status, the spouse's employment status in base year, the spouse's earnings in base year, the spouse's age and age squared, the number of children four years before a job loss, predisplacement plant size, pre-displacement region (21 categories) and industry dummies (10 categories), and time dummies*base year dummies interactions.

Also, if displacements were truly exogenous in this context, which the authors argue is the case, the effects should be apparent in the graphical analysis of means (figures 5 and 6). To my eye, these figures do not provide very convincing evidence in support of the regression-based results.

Regression Specification

Though I understand the need to economize on space, I would much prefer to see the authors begin with a simple regression model without controls. Along the lines of my comment above, if displacements are truly exogenous, then the inclusion of controls should not affect the estimated effects. It would be nice to see that this is indeed the case. Or if the authors want to instead make the case that displacements are exogenous conditional on x , y , and z , then it would be nice to start with results that control only for x , y , and z , before adding all of the control variables that they have at their disposal.

Standard errors

I could not find any discussion of how the standard error estimates were obtained.

Writeup

There is a lot of room for the writeup to be improved. I found approximately one typo per paragraph. The structure could be improved. In particular, I think that it's essential that the authors identify the effects of job loss on employment, income, and divorce; however, the amount of time that the authors spend in discussing these results takes away from their main results which focus on fertility.

Minor comments

- Why restrict the sample to a third of the universe? How much harder would it be to use the entire universe?
- Though many studies have found evidence that runs against the theoretical predictions of the quantity-quality model of fertility, it should probably be mentioned in the theoretical framework, at least as a footnote.
- There are a number of recent quasi-experimental studies that have looked at how household finances affect fertility decisions that the authors might want to mention to better put their study into context. In particular, see Melissa Kearney's recent work, Lovenheim and Mumford's paper on housing wealth, Black et al's study of the coal boom and bust, and Schaller's study of gender-specific wage shocks. Lindo's study of the effects of job loss on infant health may also be relevant. Also, the Amialchuk reference on page 7 needs to be updated.
- The commas in tables 1 and 2 need to be replaced by decimal points.
- The resolution of the figures makes them hard to read.
- I don't know what the authors are trying to convey with the last sentence of the first paragraph.